

Practical Lab Report

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Question 1: Swap Two Numbers (Menu-Driven)

Step 1 - Problem Understanding

Write a menu-driven C program to input two numbers and swap them using a third variable & without using a third variable. Also display the values before and after swapping.

Required Input(s): Choice (int), Number 1 'a' (float), Number 2 'b' (float)

Required Output: Swapped values of 'a' and 'b' with clear confirmation before and after operation

Formulas / Logic:

- Case 1 (With 3rd variable): temp = a; a = b; b = temp;
- Case 2 (Without 3rd variable): a = a + b; b = a - b; a = a - b;

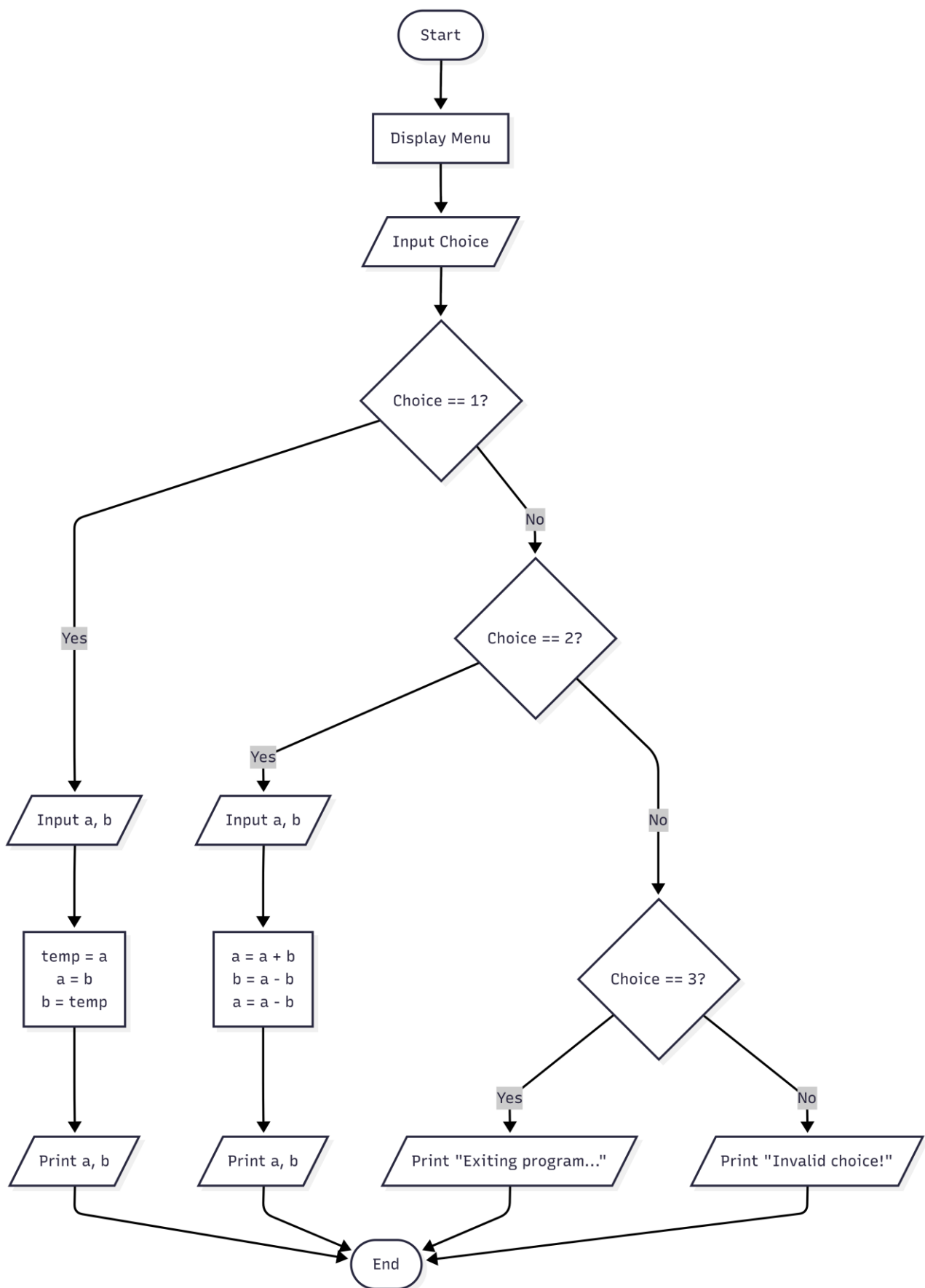
Step 2 - Test Case Table

Test Case	Choice / Inputs	Expected Output	Actual Output	Result
1	Choice = 1, a = 10, b = 20	Before: a=10.00, b=20.00 After: a=20.00, b=10.00	Before: a=10.00, b=20.00 After: a=20.00, b=10.00	Pass
2	Choice = 2, a = 5, b = 15	Before: a=5.00, b=15.00 After: a=15.00, b=5.00	Before: a=5.00, b=15.00 After: a=15.00, b=5.00	Pass
3	Choice = 3	Exiting program...	Exiting program...	Pass

Step 3 - Algorithm

1. Start.
2. Display menu options (1: Swap with 3rd var, 2: Swap without 3rd var, 3: Exit).
3. Read the user's menu choice.
4. If Choice == 1: Input 'a' and 'b'. Store temp = a, a = b, b = temp. Print results before and after.
5. If Choice == 2: Input 'a' and 'b'. Store a = a + b, b = a - b, a = a - b. Print results before and after.
6. If Choice == 3: Print exit message and terminate.
7. Stop.

Step 4 - Flowchart



Step 5 - C Program Source Code

```
#include <stdio.h>

int main() {
    int choice;
    float a, b, temp;

    printf("--- MENU ---\n");
    printf("1. Swap using third variable\n");
    printf("2. Swap without using third variable\n");
    printf("3. Exit\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            printf("Enter two numbers (a and b): ");
            scanf("%f %f", &a, &b);
            printf("Before swapping: a = %.2f, b = %.2f\n", a, b);
            temp = a;
            a = b;
            b = temp;
            printf("After swapping: a = %.2f, b = %.2f\n", a, b);
            break;
        case 2:
            printf("Enter two numbers (a and b): ");
            scanf("%f %f", &a, &b);
            printf("Before swapping: a = %.2f, b = %.2f\n", a, b);
            a = a + b;
            b = a - b;
            a = a - b;
            printf("After swapping: a = %.2f, b = %.2f\n", a, b);
            break;
        case 3:
            printf("Exiting program...\n");
            break;
        default:
            printf("Invalid choice!\n");
    }
    return 0;
}
```

Step 6 - Compilation, Execution & Evaluation

The program was compiled with GCC (gcc q1_swap.c -o q1.exe) with no errors or warnings, and executed using the inputs from Step 2.

```
PS C:\Users\Rudra Pratap Singh> gcc q1_swap.c -o q1.exe
PS C:\Users\Rudra Pratap Singh> ./q1.exe
--- MENU ---
1. Swap using third variable
2. Swap without using third variable
3. Exit
Enter choice: 1
Enter two numbers (a and b): 10 20
Before swapping: a = 10.00, b = 20.00
```

After swapping: a = 20.00, b = 10.00

```
PS C:\Users\Rudra Pratap Singh> ./q1.exe
```

```
Enter choice: 2
```

```
Enter two numbers (a and b): 5 15
```

```
Before swapping: a = 5.00, b = 15.00
```

```
After swapping: a = 15.00, b = 5.00
```

Question 2: Sum of Natural Numbers & Squares (Menu-Driven)

Step 1 - Problem Understanding

Write a menu-driven C program to find the sum of the first n natural numbers and sum of squares of the first n natural numbers.

Required Input(s): Choice (int), Natural Number 'n' (int)

Required Output: Sum of first n natural numbers OR sum of squares of first n natural numbers

Formulas / Logic:

- $\text{Sum} = n * (n + 1) / 2$
- $\text{Sum of Squares} = n * (n + 1) * (2n + 1) / 6$

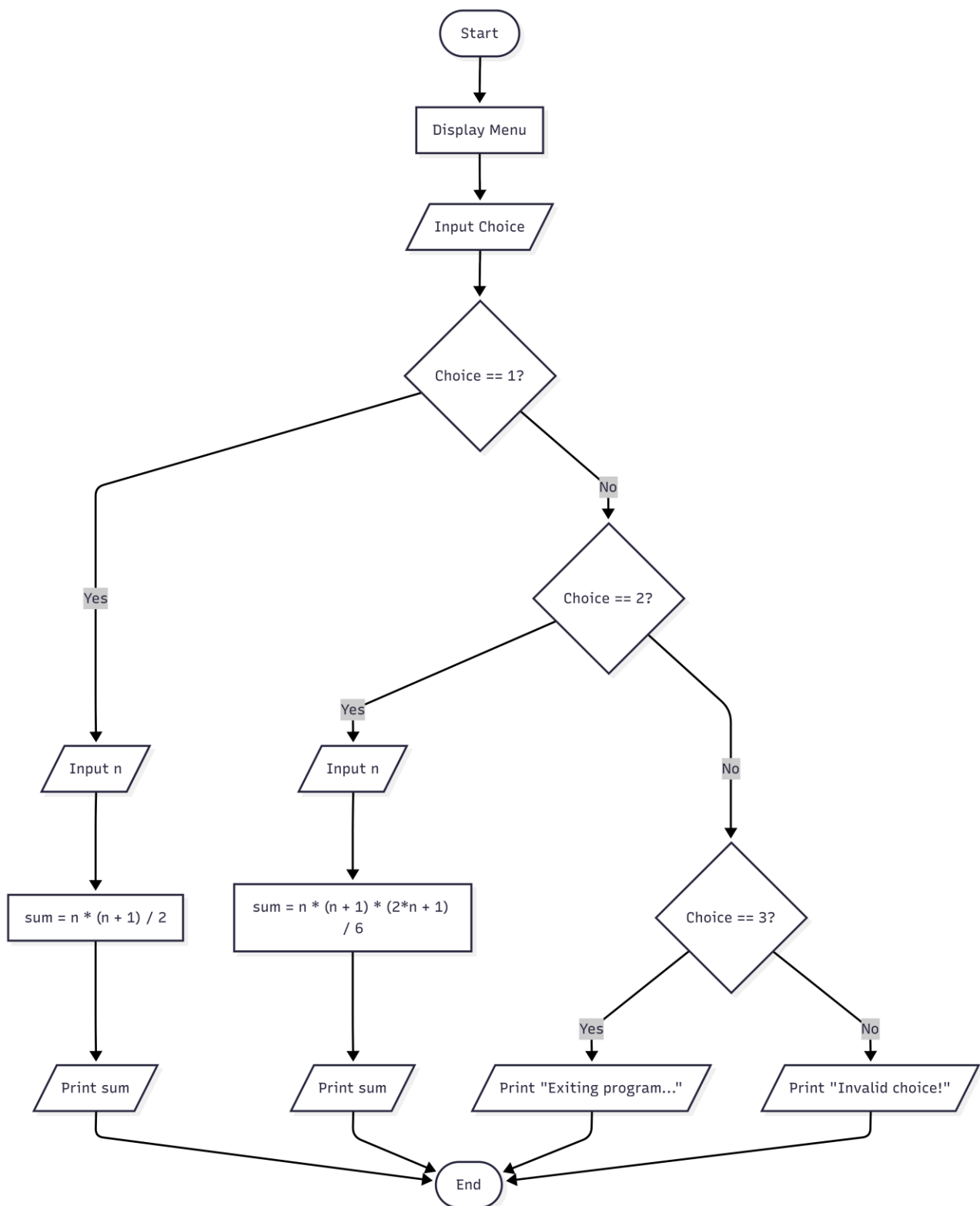
Step 2 - Test Case Table

Test Case	Choice / Inputs	Expected Output	Actual Output	Result
1	Choice = 1, n = 10	Sum of first 10 natural numbers = 55	Sum of first 10 natural numbers = 55	Pass
2	Choice = 2, n = 5	Sum of squares of first 5 natural numbers = 55	Sum of squares of first 5 natural numbers = 55	Pass
3	Choice = 3	Exiting program...	Exiting program...	Pass

Step 3 - Algorithm

1. Start.
2. Display menu options (1: Sum of n numbers, 2: Sum of squares of n numbers, 3: Exit).
3. Read choice.
4. If Choice == 1: Input 'n'. Compute $\text{sum} = n * (n + 1) / 2$. Display sum.
5. If Choice == 2: Input 'n'. Compute $\text{sum_sq} = n * (n + 1) * (2n + 1) / 6$. Display sum_sq.
6. If Choice == 3: Print exit message and stop.
7. Stop.

Step 4 – Flowchart



Step 5 - C Program Source Code

```
#include <stdio.h>
```

```

int main() {
    int choice, n, sum;

    printf("--- MENU ---\n");
    printf("1. Sum of first n natural numbers\n");
    printf("2. Sum of squares of first n natural numbers\n");
    printf("3. Exit\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            printf("Enter n: ");
            scanf("%d", &n);
            sum = n * (n + 1) / 2;
            printf("Sum of first %d natural numbers = %d\n", n, sum);
            break;
        case 2:
            printf("Enter n: ");
            scanf("%d", &n);
            sum = n * (n + 1) * (2 * n + 1) / 6;
            printf("Sum of squares of first %d natural numbers = %d\n", n, sum);
            break;
        case 3:
            printf("Exiting program...\n");
            break;
        default:
            printf("Invalid choice!\n");
    }
    return 0;
}

```

Step 6 - Compilation, Execution & Evaluation

The program was compiled with GCC (gcc q2_sum.c -o q2.exe) and tested against all test cases with exact matches.

```

PS C:\Users\Rudra Pratap Singh> gcc q2_sum.c -o q2.exe
PS C:\Users\Rudra Pratap Singh> ./q2.exe
--- MENU ---
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter choice: 1
Enter n: 10
Sum of first 10 natural numbers = 55

PS C:\Users\Rudra Pratap Singh> ./q2.exe
Enter choice: 2
Enter n: 5
Sum of squares of first 5 natural numbers = 55

```

Question 3: Financial Calculations - Simple & Compound Interest (Menu-Driven)

Step 1 - Problem Understanding

Write a menu-driven C program to calculate simple interest, compound amount, compound interest, and total payable amount for given principal, rate, and time.

Required Input(s): Choice (int), Principal P (float), Rate R (float), Time T (float)

Required Output: Simple Interest & Total Amount OR Compound Amount & Compound Interest

Formulas / Logic:

- Simple Interest (SI) = $(P * R * T) / 100$
- Total Payable Amount = $P + SI$
- Compound Amount (A) = $P * (1 + R / 100)^T$
- Compound Interest (CI) = $A - P$

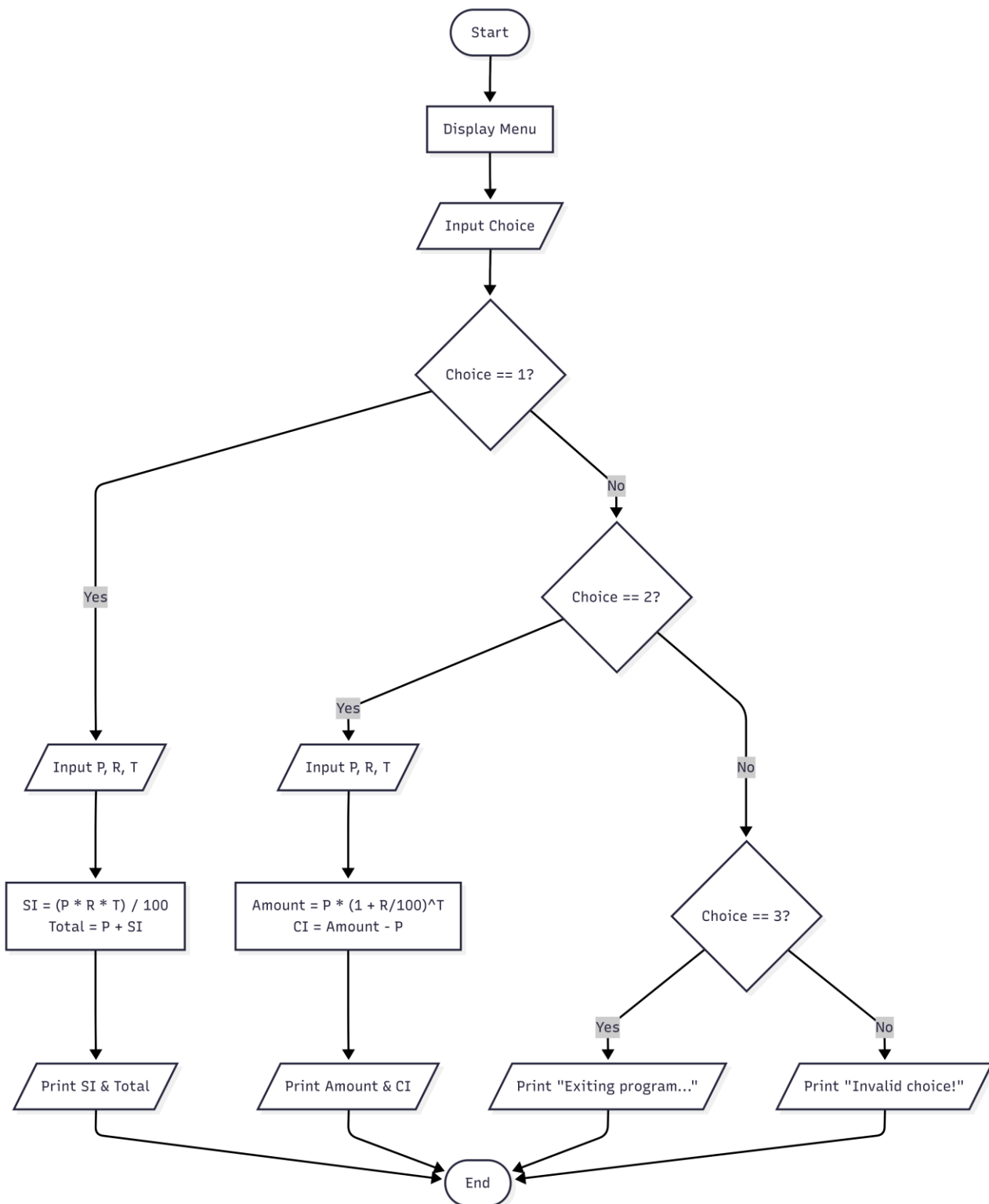
Step 2 - Test Case Table

Test Case	Choice / Inputs	Expected Output	Actual Output	Result
1	Choice = 1, P=1000, R=5, T=2	Simple Interest = 100.00 Total Payable Amount = 1100.00	Simple Interest = 100.00 Total Payable Amount = 1100.00	Pass
2	Choice = 2, P=1000, R=5, T=2	Compound Amount = 1102.50 Compound Interest = 102.50	Compound Amount = 1102.50 Compound Interest = 102.50	Pass
3	Choice = 3	Exiting program...	Exiting program...	Pass

Step 3 - Algorithm

1. Start.
2. Display menu options (1: Simple Interest & Total, 2: Compound Amount & CI, 3: Exit).
3. Read choice.
4. If Choice == 1: Input P, R, T. Compute $SI = (P * R * T) / 100$ and $Total = P + SI$. Display SI and Total.
5. If Choice == 2: Input P, R, T. Compute $Amount = P * (1 + R/100)^T$ and $CI = Amount - P$. Display Amount and CI.
6. If Choice == 3: Print exit message and terminate.
7. Stop.

Step 4 - Flowchart



Step 5 - C Program Source Code

```
#include <stdio.h>
#include <math.h>

int main() {
    int choice;
```



```

float P, R, T, SI, CI, Amount, Total;

printf("--- MENU ---\n");
printf("1. Calculate Simple Interest & Total Amount\n");
printf("2. Calculate Compound Amount & Compound Interest\n");
printf("3. Exit\n");
printf("Enter choice: ");
scanf("%d", &choice);

switch(choice) {
    case 1:
        printf("Enter Principal, Rate, Time: ");
        scanf("%f %f %f", &P, &R, &T);
        SI = (P * R * T) / 100.0;
        Total = P + SI;
        printf("Simple Interest = %.2f\n", SI);
        printf("Total Payable Amount = %.2f\n", Total);
        break;
    case 2:
        printf("Enter Principal, Rate, Time: ");
        scanf("%f %f %f", &P, &R, &T);
        Amount = P * pow((1 + R / 100.0), T);
        CI = Amount - P;
        printf("Compound Amount = %.2f\n", Amount);
        printf("Compound Interest = %.2f\n", CI);
        break;
    case 3:
        printf("Exiting program...\n");
        break;
    default:
        printf("Invalid choice!\n");
}
return 0;
}

```

Step 6 - Compilation, Execution & Evaluation

The program was compiled with GCC (gcc q3_interest.c -o q3.exe -lm) and verified against all test cases.

```

PS C:\Users\Rudra Pratap Singh> gcc q3_interest.c -o q3.exe -lm
PS C:\Users\Rudra Pratap Singh> ./q3.exe
--- MENU ---
1. Calculate Simple Interest & Total Amount
2. Calculate Compound Amount & Compound Interest
3. Exit
Enter choice: 1
Enter Principal, Rate, Time: 1000 5 2
Simple Interest = 100.00
Total Payable Amount = 1100.00

PS C:\Users\Rudra Pratap Singh> ./q3.exe
Enter choice: 2
Enter Principal, Rate, Time: 1000 5 2
Compound Amount = 1102.50
Compound Interest = 102.50

```

(Note – Theory is by AI , to be exact)